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ROLL NO- CSB24021

```
#include <stdio.h>

#include <time.h>


void constantTime() {

   int x = 10, y = 20;

   int z = x + y;   // O(1)

}


void linearTime(int n) {

   for(int i = 0; i < n; i++) {   // O(n)

      int x = i;

   }

}

void quadraticTime(int n) {

   for(int i = 0; i < n; i++) {       // O(n^2)

      for(int j = 0; j < n; j++) {

         int x = i + j;

      }

   }

}

int main() {

   int n;

   clock_t start, end;

   double time_taken;
```

```
for(n = 1000; n <= 5000; n += 1000) {
   printf("\nInput size: %d\n", n);


   start = clock();
   constantTime();
   end = clock();
   time_taken = (double)(end - start) / CLOCKS_PER_SEC;
   printf("Constant Time: %f sec\n", time_taken);


   start = clok();
   linearTime(n);
   end = clock();
   time_taken = (double)(end - start) / CLOCKS_PER_SEC;
   printf("Linear Time: %f sec\n", time_taken);


   start = clock();
   quadraticTime(n);
   end = clock();
   time_taken = (double)(end - start) / CLOCKS_PER_SEC;
   printf("Quadratic Time: %f sec\n", time_taken);
}
return 0;
```

```
Input size: 1000
Constant Time: 0.000002 sec
Linear Time: 0.000002 sec
Quadratic Time: 0.001206 sec

Input size: 2000
Constant Time: 0.000001 sec
Linear Time: 0.000003 sec
Quadratic Time: 0.004019 sec

Input size: 3000
Constant Time: 0.000001 sec
Linear Time: 0.000004 sec
Quadratic Time: 0.009345 sec

Input size: 4000
Constant Time: 0.000001 sec
Linear Time: 0.000005 sec
Quadratic Time: 0.016139 sec

Input size: 5000
Constant Time: 0.000001 sec
Linear Time: 0.000005 sec
Quadratic Time: 0.025327 sec
```

}